



Onesol Tech Hotel

SaaS Expansion & Feature Roadmap

- Multi-Tenant SaaS Architecture
- Dynamic Department Builder
- Per-Department Live Dashboards
- Google Maps & Taxi Integration
- AI Conversation Intelligence
- Guest Portal & Pre-Arrival Flow

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Phase 1 — Multi-Tenant SaaS Foundation

Super Admin controls all tenants · Each hotel client gets an isolated workspace

Super Admin Capabilities

Capability	Detail
Tenant Management	Create, suspend, or delete client hotel accounts
Billing & Subscriptions	Monthly / yearly plans, usage metering, revenue tracking
System Health Monitor	Uptime per tenant, API call volume, error rates across all clients
Impersonation	Log in as any client admin for support without sharing passwords
Global Announcements	Push notices or maintenance alerts to all clients simultaneously
White-Label Config	Custom domain, logo, and colour scheme per tenant

Client Admin Capabilities

Capability	Detail
Property Setup	Hotel name, address, logo, timezone, WhatsApp number binding
Department Builder	Create any department, name it, define custom fields (see Phase 2)
Staff Accounts	Create logins for managers and staff with granular role assignment
WhatsApp Config	Plug in their own Green API credentials — fully isolated
AI Persona Config	Name the bot, set tone/voice, upload hotel info and FAQs
Billing Portal	View plan, usage counters, download invoices

Role Hierarchy

Four tiers of access, each scoped to exactly the data and actions they need:

- Super Admin — full platform control, all tenants, billing, system health
- Client Admin — single property or hotel group, full config + all operational data
- Department Manager — their department only: requests, staff, analytics, config
- Staff / Operator — view and action requests only, no configuration access

Phase 2 — Dynamic Department Builder

The biggest differentiator — any department, any fields, any workflow

Today the four departments (Housekeeping, Room Service, Concierge, Maintenance) are hardcoded. The Department Builder replaces this entirely. Client admins design their own service structure.

How It Works

- Step 1: Client admin opens Department Builder !' clicks '+ New Department'
- Step 2: Names the department (e.g. Pool Services, Valet, Event Management, Spa Bookings)
- Step 3: Defines custom fields for requests: Text, Number, Dropdown, Time Picker, Toggle
- Step 4: Saves — the department immediately appears everywhere in the system

Where New Departments Automatically Appear

Location	What Happens
WhatsApp Bot Menu	Auto-numbered menu option added, keyword routing configured
Requests Table	Custom columns appear matching the fields defined
Per-Dept Dashboard	A dedicated live dashboard created for that department
Analytics Page	Department included in all charts and breakdowns
Staff Assignment	Staff can be assigned to the new department immediately

Per-Department Dashboard — What Staff See

Every department gets a private real-time view. Only that department's team has access:

- Open / Active / Completed request counts — live counters
- Average resolution time for the department
- Request queue with custom field values visible inline
- Accept !' Assign to Staff !' In Progress !' Complete workflow buttons
- Each status change fires a WhatsApp update back to the guest automatically
- Filter by date, status, staff member, or custom field values

Phase 3 — Google Maps & Taxi Integration

Real location intelligence across the entire guest journey

WhatsApp Taxi Booking Flow

When a guest requests a taxi, the bot collects pickup time and location, confirms the booking, and sends a live Google Maps tracking link directly in WhatsApp. No app install required by the guest.

Google Maps Feature Set

Feature	Implementation
Live Taxi Tracking	Embed Maps link with driver position shared to guest via WhatsApp
ETA Calculation	Google Directions API ! 'Approx 22 minutes to airport'
Hotel Location Card	Auto-send hotel pin to guest on check-in confirmation
Nearby POI	Pharmacy, restaurants, ATMs with distance and rating
Airport Distance	Auto-fill in taxi booking confirmation message
Indoor Hotel Map	Upload floor plan — guests navigate to restaurant/gym/spa
Pre-Arrival Directions	Send guests a directions link 24h before arrival
Concierge Dispatch Map	Live map in dashboard showing all active pickups and ETAs

Taxi Dispatch Dashboard (Concierge Department View)

- Live map showing all active guest pickups with real-time positions
- Driver assignment — integrate with Uber for Business API or local taxi providers
- ETA countdown per booking with guest notification automation
- Guest alerts: Driver assigned !' En route !' Arrived at lobby
- Full booking log with guest name, room, destination, time, fare

Phase 4 — AI & Conversation Intelligence

Move from reactive responses to proactive, personalised guest relationships

Feature	Detail
Guest Memory Across Stays	Remember returning guests' preferences: pillow type, floor, dietary needs
Proactive Messaging	'Good morning — checkout is at 12pm. Need a late checkout?'
Pre-Arrival Sequence	Auto-message 24h before: 'Anything we should prepare for your stay?'
Upselling Engine	Detect room type !' offer relevant upgrades with price
Sentiment Detection	Flag conversations where guest tone turns negative !' alert manager
Human Escalation	Detect frustration !' 'Let me connect you with our manager' !' human takeover
Voice Note Transcription	Guest sends WhatsApp voice note !' transcribed + processed by AI
Image Support	Guest sends photo of broken item !' maintenance request auto-created
Multi-Language	Auto-detect guest language and reply in the same language
Post-Stay Survey	Auto-send NPS survey on checkout, results feed into dashboard

Phase 5 — Operations Dashboard Upgrades

From conversation metrics to full hotel operations visibility

New Dashboard Pages

Page	Content
Room Map	Visual floor-plan grid — each room shows status: Occupied / Vacant / Dirty / Maintenance
Revenue	RevPAR, ADR, occupancy %, ancillary revenue — today vs. last week vs. last month
Guest Profiles	Searchable list — stay history, preferences, request history, sentiment score
Live Map	Google Maps view with all active taxis, hotel location, guest-shared POIs
Escalations	Queue of conversations flagged for human takeover — reply directly from dashboard
Notifications	Rule-based alerts: open request >30min, negative sentiment, >5 pending requests

New KPIs to Track

- NPS / Satisfaction Score — auto-survey sent on checkout via WhatsApp
- Request resolution time per department
- Upsell conversion rate (offers sent vs. accepted)
- Returning guest rate
- Bot escalation rate — % of conversations needing human intervention
- Revenue per conversation (when booking/order data is tracked)

Phase 6 — Guest Self-Service Portal

A lightweight web app — no download, sent as a link on check-in

Guests receive a personalised link when they check in. It opens a mobile-optimised web app authenticated by a UUID in the URL — no password needed. Everything they need in one place.

Portal Features

Feature	Detail
Request Service	Trigger any department request — fields match whatever was configured
Order Room Service	Full menu with add-to-cart, special instructions, order tracking
Book Taxi	Select time and destination — live tracking link returned
Book Spa / Facilities	View availability, select slot, instant confirmation to WhatsApp
Hotel Map	Interactive indoor map — navigate to restaurant, gym, pool, rooms
Local Recommendations	AI-curated nearby POIs based on guest profile and stay dates
View My Bill	Running total of room service and service charges
Chat with Alex	Opens WhatsApp conversation with the AI receptionist

Phase 7 — Integrations & Ecosystem

Connect to the tools hotels and guests already use

Integration	Value
Google Maps API	Taxi tracking, POI discovery, directions, indoor hotel mapping
Uber for Business / Careem	Real dispatched taxi booking with driver assignment and tracking
Stripe / Paddle	SaaS subscription billing — plans, usage metering, invoices
Twilio	SMS fallback when WhatsApp is unavailable or undelivered
SendGrid	Transactional emails — booking confirmations, invoices, surveys
Slack	Push escalation alerts directly to hotel manager's Slack channel
Zapier / Make	Let clients connect to any tool without writing code
Google Reviews API	Pull review scores and sentiment into the dashboard automatically
PMS (Cloudbeds / Mews)	Sync room status, bookings, and guest data bidirectionally
Payment Terminals	Accept card payment for room service via saved card on file

Phase 8 — Monetisation & SaaS Tiers

Four plans covering every hotel size from boutique to enterprise chain

Plan	Target	Price	Key Features
Starter	Small hotel, 1 property	\$49/mo	1 dept, 500 conv/mo, basic dashboard
Professional	Mid-size, 1 property	\$149/mo	Unlimited depts, 5K conv, Maps, profiles
Business	Hotel chain, multi-property	\$399/mo	Multi-property, white-label, API access
Enterprise	Large chains	Custom	Dedicated DB, SLA, custom integrations

Upsell revenue streams beyond subscriptions: per-conversation overage charges, SMS/voice credits, Google Maps API pass-through, white-label licensing fee for resellers.

Appendix — Priority Build Order

Recommended implementation sequence based on dependencies and business impact

1.	Multi-tenant auth system	Foundation
2.	Department Builder with custom fields	Differentiator
3.	Per-department dashboards	Immediate value
4.	Database + guest profiles persistence	Required for memory
5.	Google Maps taxi flow	Guest-facing
6.	Human escalation (dashboard takeover)	Critical gap
7.	Pre-arrival & proactive messaging	Revenue driver
8.	Guest portal web app	Self-service
9.	Revenue & analytics dashboards	GM-level value
10.	SaaS billing & subscription management	Monetisation

The current system is already ahead of most hotel chatbot SaaS tools in AI quality and real-time operations visibility. The SaaS layer, Department Builder, Google Maps, and Guest Portal would make this a genuinely competitive product in the \$49–\$399/month market.